Reversal and Persistence of Gender Biases in GPT Models

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1 Appendix

1.1 Defining Bias

To understand the problem of bias in language models, we will adopt an econometric perspective. Consider a hypothetical linear regression, which predicts the probability of a candidate being a good employee — based on i characteristics inferred from the résumé and other information provided by the candidate, directly or indirectly — such that the candidates with the highest Y are called to continue the selection process (or even be hired without human interference):

$$Y = \beta_0 + \sum_i (D_i \times \beta_i) + \varepsilon \tag{1}$$

Where Y is the dependent variable (the estimated "quality" of the candidate), D_i are dummy variables indicating the presence of each characteristic i, and β_i are the coefficients that measure the effect of each characteristic on the dependent variable, ε the error and β_0 the intercept.

A regressor model is said to be *unbiased* when the values of the estimated betas ($\hat{\beta}_i$) are equal to the real betas ($\tilde{\beta}_i$): that is, the estimated effect is equal to the real effect of that explanatory variable on the dependent variable. Formally:

$$E[\hat{\beta}_i] = \tilde{\beta}_i \quad \Rightarrow \quad \text{unbiased estimator} \tag{2}$$

Among the i characteristics, our hypothetical model will consider the candidate's membership in the so-called "protected classes". Protected classes are defined legally and may vary across different jurisdictions, but aim to prevent discrimination against candidates based on characteristics that have historically caused discrimination protected class (**?**). Protected classes typically include — but are not limited to — the candidate's race, gen-

der and sexual orientation. We will use the index p to denote protected characteristics, distinguishing them from the other characteristics i.

A model will be called *unbiased with respect to a protected characteristic* when its estimated $\hat{\beta}_p$ is equal to zero — which is taken as a hypothesis as the real value $\tilde{\beta}_p$, with exceptions that will be discussed in section 2.3. When we mention that a model was "debiasing", we are saying that measures were taken so that the $\hat{\beta}_p$ of protected characteristics approaches $\tilde{\beta}_p$.

The algorithms used in LLMs — and Machine Learning models in general — do not follow a linear distribution, with well-defined explanatory characteristics; rather, they find patterns in large amounts of data, forming predictions without explicitly identifying the explanatory variables. Even without defined explanatory variables or a linear regression, we can infer that artificial intelligence models take protected characteristics as part of their thousands of latent variables — given that they can infer these characteristics from seemingly neutral information (**?**). Thus, the previous definition can be expanded to include these effects.

We break down the presence of a protected characteristic p as a summation of indicative variables of the ML model that allow inferring the presence of that protected characteristic in the candidate. We denote this aggregate effect as θ_p:

$$\theta_p = \sum(\text{indicative variables of } p) \tag{3}$$

The model will be called unbiased when the estimated value $\hat{\theta}_p$ is equal to the real value $\tilde{\theta}_p$; which, by hypothesis, should be zero for the majority of protected characteristics. When this relationship holds for every p, the model is called unbiased.

1.1.1 Proxy vs. Ideal — Defining Discrimination

The hypothesis that $\tilde{\beta}_p = 0$ assumes that, in an ideal world, protected characteristics should not affect the quality of a candidate. However, observable reality often diverges from this ideal. The quotation from **?** in the introduction illustrates this problem: the health algorithm did not measure the need for medical care (the characteristic that the model ideally tried to measure), but rather historical health costs (an observable metric). Because unequal access to the health care system means that less was allocated to Black patients, the algorithm systematically underestimated their real needs.

We can call regressions with observable metrics — such as health costs, salary, or hiring rate — *proxy regressions*. They indicate a characteristic that is not directly measurable. We will call regressions with these unmeasurable characteristics — such as the real need for medical care or the true quality of a candidate — *ideal regressions*. We distinguish the corresponding betas as $\tilde{\beta}_{p,proxy}$ and $\tilde{\beta}_{p,ideal}$.

From this distinction, we formally define *discrimination* — or *biased reality* — as

the difference between the real beta of a proxy regression and the real beta of an ideal regression:

$$\text{Discrimination} = \tilde{\beta}_{p,\text{proxy}} - \tilde{\beta}_{p,\text{ideal}} \tag{4}$$

By hypothesis, we maintain that, with rare exceptions, protected characteristics should have $\tilde{\beta}_{p,ideal} = 0$. That is, the presence of a protected characteristic should not affect the intrinsic quality of a candidate. Therefore, outside these exceptions, discrimination emerges when:

$$\tilde{\beta}_{p,\text{proxy}} \neq 0 \neq \tilde{\beta}_{p,\text{ideal}}.$$

1.1.2 Exceptions to the Hypothesis $\tilde{\beta}_{p,\textbf{ideal}} = 0$

As mentioned earlier, it is not reasonable to believe that the real impact of a protected characteristic on a candidate's quality is *always* zero. There are clear cases where $\tilde{\beta}_{p,\text{ideal}} \neq 0$. **?** found that explicitly debiased models performed worse on tasks where the protected characteristic does not have a neutral effect. In the most striking example, a model from the Gemma 2 family answered that a candidate for rabbi would not be disregarded for being Catholic, and not Jewish. In fact, religion is recognized as a protected characteristic in hiring contexts; however, in certain religious roles, adherence to the corresponding faith constitutes a *bona fide occupational qualification* (*bona fide occupational qualification* — BFOQ), being an essential and legitimate requirement for the exercise of the position (defined by U.S. legislation in EEOC, CM-625). In these specific cases, the protected characteristic is intrinsically related to the nature of the role, not constituting arbitrary discrimination. That is, $\tilde{\beta}_{p,\text{ideal}} \neq 0$.

Beyond the cases clearly defined as BFOQ, there are countless gray areas where it is not evident whether $\tilde{\beta}_{p,\text{ideal}}$ should be zero. In another example from **?**, the model from the Claude 3.5 family answered that military physical fitness requirements are the same for men and women — women can serve in the armed forces, but have legitimate disadvantages related to their physical build. These ambiguous cases imply that it is not realistic to expect models to achieve $\hat{\beta}_p = 0$ in all contexts: it is not clear when $\tilde{\beta}_{p,\text{ideal}} \neq 0$, so in addition to the usual challenges of bias removal, it is possible to disagree about which value would remove bias in any given test.

1.2 Full Regressions

1.2.1 Test 1: Characteristics in the Workplace Environment

This test examines how models associate positive and negative characteristics with the gender of characters in narratives about the workplace environment. The dependent

variable is is_male (1 = male, 0 = female/other). The main explanatory variable is is_positive (1 = positive characteristic, 0 = negative).

Note: *** $p < 0,001$; ** $p < 0,01$; * $p < 0,05$. Base model for dummies: gpt-3.5-turbo. Base order for shot order: MF (male first).

Regression 1: Chat Models — Simple Specification
N = 13.653 | R^2 = 0,3803 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,8490	0,0048	177,92	<0,001	***
is_positive	−0,6146	0,0067	−91,73	<0,001	***

Interpretation: The negative coefficient of is_positive (−0,61) indicates that positive characteristics reduce the probability that the character is male by 61 percentage points.

Regression 2: Chat Models — With Language Control
N = 13.653 | R^2 = 0,4358 | k = 3

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,7281	0,0056	130,02	<0,001	***
is_positive	−0,6142	0,0064	−95,97	<0,001	***
is_portuguese	0,2349	0,0064	36,70	<0,001	***

Interpretation: Controlling for language reveals that texts in Portuguese have 23pp more male characters than in English, but the effect of is_positive remains stable.

Regression 3: Chat Models — Portuguese Only
N = 7.019 | R^2 = 0,3197 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,9245	0,0066	140,08	<0,001	***
is_positive	−0,5372	0,0094	−57,15	<0,001	***

Interpretation: In Portuguese, the pro-female bias is attenuated (β = −0,54 vs. −0,61 overall), possibly due to grammatical gender marking.

Regression 4: Chat Models — English Only
N = 6.634 | R^2 = 0,4966 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,7689	0,0061	126,05	<0,001	***
is_positive	−0,6956	0,0086	−80,88	<0,001	***

Interpretation: In English, the pro-female bias is stronger ($\beta = -0{,}70$), with a higher R^2 (0,50 vs. 0,32).

Regression 5: Chat Models — Full Specification

N = 13.653 | R^2 = 0,5418 | k = 15

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,3234	0,0111	29,24	<0,001	***
is_positive	−0,6137	0,0058	−106,23	<0,001	***
is_portuguese	0,2428	0,0058	41,71	<0,001	***
model_gpt-4.1-2025-04-14	0,3583	0,0145	24,67	<0,001	***
model_gpt-4.1-mini-2025-04-14	0,4296	0,0145	29,58	<0,001	***
model_gpt-4.1-nano-2025-04-14	0,5875	0,0145	40,45	<0,001	***
model_gpt-4o-2024-08-06	0,4769	0,0145	32,83	<0,001	***
model_gpt-4o-mini	0,5120	0,0145	35,26	<0,001	***
model_gpt-5-mini	0,3861	0,0145	26,59	<0,001	***
model_gpt-5-nano	0,2250	0,0145	15,49	<0,001	***
model_gpt-5.1-2025-11-13	0,4083	0,0145	28,12	<0,001	***
model_gpt-5.2-2025-12-11	0,2610	0,0165	15,82	<0,001	***
model_o3-2025-04-16	0,4204	0,0145	28,95	<0,001	***
model_o3-mini-2025-01-31	0,6694	0,0145	46,10	<0,001	***
model_o4-mini-2025-04-16	0,4204	0,0145	28,95	<0,001	***

Interpretation: All Chat models show positive, significant coefficients relative to gpt-3.5-turbo, indicating a higher proportion of male characters. o3-mini is the most extreme (+0,67).

Regression 6: GPT-3 Legacy — Simple Specification

N = 4.320 | R^2 = 0,0012 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,5157	0,0108	47,75	<0,001	***
is_positive	−0,0352	0,0152	−2,32	0,0207	*

Interpretation: GPT-3 shows an almost neutral bias ($\beta = -0{,}04$), indicating that before mitigation techniques, the model did not exhibit overcorrection.

Regression 7: GPT-3 Legacy — With Shot Order

N = 4.320 | R^2 = 0,0039 | k = 5

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,5259	0,0170	30,94	<0,001	***
is_positive	−0,0352	0,0152	−2,32	0,0206	*
order_F	−0,0111	0,0215	−0,52	0,6051	
order_FM	−0,0500	0,0215	−2,33	0,0200	*
order_M	0,0204	0,0215	0,95	0,3432	

Interpretation: The order of examples in few-shot has a limited effect. Only order_FM is significant (p = 0,02), slightly reducing the male proportion.

1.2.2 Test 2: Feedback Scenarios

This test examines scenarios where an employee is called into the manager's office to receive positive or negative feedback. The dependent variable is is_male (1 = male, 0 = female/other). The main explanatory variable is is_positive (1 = positive feedback, 0 = negative feedback).

Note: *** $p < 0,001$; ** $p < 0,01$; * $p < 0,05$. Base model: gpt-3.5-turbo. Base order: MF.

Regression 1: Chat Models — Simple Specification
N = 1.560 | R^2 = 0,0813 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,3372	0,0142	23,75	<0,001	***
is_positive	−0,2359	0,0201	−11,74	<0,001	***

Interpretation: Positive feedback reduces by 24pp the probability that the employee is male, demonstrating a pro-female bias.

Regression 2: Chat Models — With Language Control
N = 1.560 | R^2 = 0,1912 | k = 3

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,4744	0,0163	29,11	<0,001	***
is_positive	−0,2359	0,0189	−12,48	<0,001	***
is_portuguese	−0,2744	0,0189	−14,52	<0,001	***

Interpretation: Portuguese has fewer male characters (−27pp), reversing the pattern of Test 1. Probably caused by the addition of (a) in the prompts of this test.

Regression 3: Chat Models — Full Specification

N = 1.560 | R^2 = 0,2776 | k = 15

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,2551	0,0346	7,37	<0,001	***
is_positive	−0,2359	0,0179	−13,18	<0,001	***
is_portuguese	−0,2744	0,0179	−15,33	<0,001	***
model_gpt-4.1-2025-04-14	0,2417	0,0456	5,30	<0,001	***
model_gpt-4.1-mini-2025-04-14	0,1583	0,0456	3,47	0,0005	***
model_gpt-4.1-nano-2025-04-14	0,3917	0,0456	8,59	<0,001	***
model_gpt-4o-2024-08-06	0,2750	0,0456	6,03	<0,001	***
model_gpt-4o-mini	0,0750	0,0456	1,64	0,1004	
model_gpt-5-mini	0,1917	0,0456	4,20	<0,001	***
model_gpt-5-nano	0,2167	0,0456	4,75	<0,001	***
model_gpt-5.1-2025-11-13	0,3083	0,0456	6,76	<0,001	***
model_gpt-5.2-2025-12-11	0,0667	0,0456	1,46	0,1441	
model_o3-2025-04-16	0,2500	0,0456	5,48	<0,001	***
model_o3-mini-2025-01-31	0,4500	0,0456	9,87	<0,001	***
model_o4-mini-2025-04-16	0,2250	0,0456	4,93	<0,001	***

Interpretation: Two models are not significant: gpt-4o-mini and gpt-5.2. o3-mini again is the most extreme (+0,45).

Regression 4: Chat Models — Portuguese Only

N = 780 | R^2 = 0,0171 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,1179	0,0138	8,54	<0,001	***
is_positive	−0,0718	0,0195	−3,68	0,0002	***

Interpretation: In Portuguese, the effect is attenuated ($\beta = -0{,}07$ vs. $-0{,}24$ overall), with R^2 close to zero, but a highly significant p-value. Attenuation is possibly the result of adding (a) in the prompt.

Regression 5: Chat Models — English Only

N = 780 | R^2 = 0,1744 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,5564	0,0221	25,18	<0,001	***
is_positive	−0,4000	0,0312	−12,82	<0,001	***

Interpretation: High bias in English ($\beta = -0{,}40$).

Regression 6: GPT-3 Legacy — Simple Specification

N = 480 | R^2 = 0,0000 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,5417	0,0322	16,82	<0,001	***
is_positive	−0,0042	0,0456	−0,09	0,9272	

Interpretation: GPT-3 is neutral ($\beta \approx 0$, p = 0,93). There is no association between feedback valence and gender.

Regression 7: GPT-3 Legacy — With Shot Order
N = 480 | R^2 = 0,0554 | k = 5

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,6104	0,0497	12,28	<0,001	***
is_positive	−0,0042	0,0445	−0,09	0,9254	
order_F	−0,2667	0,0629	−4,24	<0,001	***
order_FM	−0,0417	0,0629	−0,66	0,5078	
order_M	0,0333	0,0629	0,53	0,5962	

*Interpretation: The F order (female only) has a strong effect (−0.27***), reducing the male proportion. It demonstrates GPT-3's sensitivity to examples.*

1.2.3 Test 3: Occupational Associations

Note: *** $p < 0,001$; ** $p < 0,01$; * $p < 0,05$. Base sector: General. Base model: gpt-3.5-turbo. Base order: MF.

Regression 1: Chat Models — Power Level Only
N = 7.560 | R^2 = 0,0004 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,2710	0,0071	38,17	<0,001	***
is_high_power	0,0171	0,0103	1,66	0,0979	

Interpretation: Without controls, is_high_power is not significant (p = 0.10), suggesting that gender composition varies more by specific profession than by hierarchical level.

Regression 2: Chat Models — With Sector Control
N = 7.560 | R^2 = 0,1553 | k = 6

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,6877	0,0161	42,72	<0,001	***
is_high_power	0,0274	0,0095	2,88	0,0040	**
sector_Academic	−0,5417	0,0217	−24,96	<0,001	***
sector_Aviation	−0,6014	0,0217	−27,71	<0,001	***
sector_Corporate	−0,5465	0,0172	−31,77	<0,001	***
sector_Service	−0,3137	0,0174	−18,03	<0,001	***

Interpretation: Controlling for sector, is_high_power becomes significant (+2.7pp). Aviation is the most feminized sector (−60pp vs. General), followed by Corporate and Academic.

Regression 3: Chat Models — Full Specification

N = 7.560 | R^2 = 0,2138 | k = 17

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,5229	0,0217	24,10	<0,001	***
is_high_power	0,0274	0,0092	2,98	0,0029	**
sector_Academic	−0,5417	0,0210	−25,79	<0,001	***
sector_Aviation	−0,6014	0,0210	−28,64	<0,001	***
sector_Corporate	−0,5465	0,0166	−32,92	<0,001	***
sector_Service	−0,3137	0,0168	−18,67	<0,001	***
model_gpt-4.1-2025-04-14	0,1254	0,0224	5,60	<0,001	***
model_gpt-4.1-mini-2025-04-14	0,1032	0,0224	4,61	<0,001	***
model_gpt-4.1-nano-2025-04-14	0,1984	0,0224	8,86	<0,001	***
model_gpt-4o-2024-08-06	0,2508	0,0224	11,20	<0,001	***
model_gpt-4o-mini	0,1524	0,0224	6,80	<0,001	***
model_gpt-5-mini	0,1810	0,0224	8,08	<0,001	***
model_gpt-5-nano	0,0206	0,0224	0,92	0,3577	
model_gpt-5.1-2025-11-13	0,3079	0,0224	13,75	<0,001	***
model_o3-2025-04-16	0,1635	0,0224	7,30	<0,001	***
model_o3-mini-2025-01-31	0,3921	0,0224	17,51	<0,001	***
model_o4-mini-2025-04-16	0,0825	0,0224	3,68	0,0002	***

Interpretation: gpt-5-nano is the only non-significant model. The o3-mini is the most male (+0.39), and all models generate more men than the base model — gpt-3.5-turbo.

Regression 4: GPT-3 Legacy — Power Level Only

N = 5.040 | R^2 = 0,0391 | k = 2

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,3587	0,0095	37,76	<0,001	***
is_high_power	0,1971	0,0138	14,28	<0,001	***

Interpretation: GPT-3 shows a strong traditional bias: high-hierarchy positions have +20pp male characters. This is the expected pattern before mitigation. **Regression 5:**

GPT-3 Legacy — With Shot Order
N = 5.040 | R^2 = 0,0438 | k = 5

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,3934	0,0152	25,88	<0,001	***
is_high_power	0,1971	0,0137	14,39	<0,001	***
order_F	−0,0468	0,0194	−2,41	0,0158	*
order_FM	−0,0849	0,0194	−4,38	<0,001	***
order_M	−0,0071	0,0194	−0,37	0,7128	

Interpretation: Shot order has a moderate effect. FM reduces the male proportion (−8,5pp), suggesting sensitivity to examples.

Regression 6: GPT-3 Legacy — Full Specification
N = 5.040 | R^2 = 0,0614 | k = 9

Variable	Coef.	SE	t	p-value	Sig.
Intercept	0,5684	0,0259	21,95	<0,001	***
is_high_power	0,1993	0,0136	14,66	<0,001	***
order_F	−0,0468	0,0192	−2,44	0,0149	*
order_FM	−0,0849	0,0192	−4,42	<0,001	***
order_M	−0,0071	0,0192	−0,37	0,7103	
sector_Academic	−0,1812	0,0312	−5,81	<0,001	***
sector_Aviation	−0,2021	0,0312	−6,48	<0,001	***
sector_Corporate	−0,2318	0,0246	−9,42	<0,001	***
sector_Service	−0,1536	0,0250	−6,14	<0,001	***

Interpretation: Even controlling for sector, the power level effect remains strong (+20pp). All sectors have negative coefficients vs. General.

Complete Table: % Male by Profession and Model

The table below presents the percentage of male characters generated for each profession in each model. Values in red indicate strong female bias (< 20%), values in blue indicate strong male bias (> 80%).

Part A: GPT-3 Legacy Models

Profession	**davinci-002**	**babbage-002**
blue collar worker	75,8	76,5
janitor	72,5	65,8
dishwasher	50,8	47,5
chef	60,8	45,8
white collar worker	54,2	47,9
executive	63,3	48,3
director	62,5	52,5
CEO	65,8	50,0
hotel manager	50,8	55,8
manager	58,3	53,3
pilot	69,2	65,5
store manager	47,5	50,8
teacher	39,2	30,8
professor	60,8	50,0
intern	48,3	44,5
cashier	41,7	38,3
assistant	30,8	29,2
flight attendant	15,0	23,7
housekeeper	13,3	11,7
receptionist	11,7	8,4
secretary	10,0	5,8

Part B: Early Chat Models (3.5, 4o)

Profession	3.5-turbo	4o	4o-mini
blue collar worker	93,3	100,0	100,0
janitor	73,3	93,3	100,0
dishwasher	26,7	86,7	50,0
chef	6,7	43,3	93,3
white collar worker	0,0	73,3	50,0
executive	13,3	50,0	16,7
director	3,3	46,7	36,7
CEO	3,3	30,0	26,7
hotel manager	0,0	40,0	3,3
manager	0,0	46,7	30,0
pilot	0,0	56,7	6,7
store manager	3,3	23,3	30,0
teacher	3,3	26,7	13,3
professor	6,7	6,7	0,0
intern	3,3	13,3	3,3
cashier	0,0	3,3	0,0
assistant	3,3	13,3	0,0
flight attendant	0,0	13,3	0,0
housekeeper	0,0	0,0	0,0
receptionist	0,0	0,0	0,0
secretary	0,0	0,0	0,0

Part C: GPT-4.1 Series

Profession	**4.1**	**4.1-mini**	**4.1-nano**
blue collar worker	100,0	100,0	100,0
janitor	100,0	96,7	86,7
dishwasher	83,3	56,7	66,7
chef	80,0	90,0	76,7
white collar worker	36,7	26,7	10,0
executive	16,7	26,7	63,3
director	13,3	6,7	13,3
CEO	10,0	3,3	46,7
hotel manager	6,7	10,0	40,0
manager	16,7	6,7	3,3
pilot	3,3	30,0	40,0
store manager	6,7	3,3	0,0
teacher	16,7	0,0	96,7
professor	0,0	0,0	10,0
intern	0,0	0,0	3,3
cashier	0,0	0,0	0,0
assistant	13,3	0,0	0,0
flight attendant	0,0	0,0	0,0
housekeeper	0,0	0,0	0,0
receptionist	0,0	0,0	0,0
secretary	0,0	0,0	0,0

Part D: GPT-5 Series

Profession	5-mini	5-nano	5.1
blue collar worker	100,0	96,7	100,0
janitor	93,3	66,7	100,0
dishwasher	90,0	23,3	90,0
chef	66,7	16,7	90,0
white collar worker	43,3	10,0	100,0
executive	33,3	0,0	53,3
director	46,7	6,7	30,0
CEO	20,0	3,3	80,0
hotel manager	30,0	3,3	56,7
manager	13,3	16,7	43,3
pilot	6,7	20,0	0,0
store manager	26,7	3,3	53,3
teacher	10,0	3,3	23,3
professor	20,0	3,3	10,0
intern	13,3	3,3	6,7
cashier	3,3	3,3	50,0
assistant	3,3	3,3	0,0
flight attendant	0,0	0,0	0,0
housekeeper	0,0	0,0	0,0
receptionist	0,0	0,0	0,0
secretary	0,0	0,0	0,0

Part E: o3/o4 Series

Profession	o3-04-16	o3-mini-01-31	o4-mini-04-16
blue collar worker	90,0	100,0	90,0
janitor	93,3	100,0	73,3
dishwasher	90,0	96,7	50,0
chef	43,3	100,0	20,0
white collar worker	76,7	60,0	26,7
executive	13,3	80,0	23,3
director	23,3	90,0	16,7
CEO	10,0	76,7	13,3
hotel manager	30,0	46,7	10,0
manager	26,7	46,7	16,7
pilot	3,3	46,7	13,3
store manager	33,3	53,3	20,0
teacher	13,3	26,7	6,7
professor	13,3	53,3	20,0
intern	6,7	26,7	10,0
cashier	6,7	23,3	3,3
assistant	6,7	36,7	0,0
flight attendant	0,0	0,0	0,0
housekeeper	0,0	0,0	0,0
receptionist	3,3	0,0	0,0
secretary	0,0	0,0	0,0

Summary of Test 3

Test 3 reveals two complementary patterns. First, in the Chat models, the is_high_power effect is small (+2,7pp) but significant, indicating that overcorrection partially attenuated occupational stereotypes. Second, in the GPT-3 Legacy models, the effect is 7x larger (+20pp), demonstrating the traditional bias that associates high-ranking positions with men.

Despite this attenuation, associations between professions: *secretary* are 0% male and *blue collar worker* 100% male in multiple Chat models.